Object Oriented Programming (CS1004)

Course Instructor(s):
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Section(s): SE (A, B, C, D)

Sessional-I Exam

Total Time (Hrs): 1
Total Marks: 60
Total Questions: 1

Date: Feb 24, 2026

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Roll No **Course Section** **Student Signature**

Do not write below this line.

Attempt all the questions.

[CLO1: Demonstrate the basic concepts of OOP]

Q1: What would be the output produced by executing the following C++ codes? Identify errors, if any (either write output or error (syntax/runtime), both will not be accepted). All the code snippet contains #include and using namespace std; & these programs are compiled using g++ on a 32-bit system architecture. **[5 + 5 + 5 + 10 + 5 + 5 + 5 + 5 + 5 + 10 = 60 marks]**

1.	<pre>struct Record { char *text; }; void manipulate(Record r, char *p) { r.text = r.text + 1; *(r.text + 1) = *(p + 3); p = p + 2; *(p) = *(p - 2); } int main() { char arr[] = {'A','B','C','D','E','F','G','\0'}; Record r1, r2; r1.text = arr; r2.text = r1.text + 3; char *p1 = r1.text + 2; char *p2 = r2.text - 1; *(p1) = *(p2 + 2); *(r2.text) = *(r1.text + 1); manipulate(r1, arr); r2.text = r1.text + 1; *(r2.text + 2) = '\0'; *(r1.text + 5) = *(r1.text); cout << r1.text << endl; cout << r2.text << endl; return 0; }</pre>	**5 marks** **ABA** **BA**

2.	```int main() { struct { int x; int *p; } arr[2]; int a = 5, b = 10; arr[0].x = 1; arr[0].p = &a; arr[1].x = 2; arr[1].p = &b; *(arr[0].p) += arr[1].x; arr[1].p = arr[0].p; *(arr[1].p) += 3; cout << a << " " << b << endl; return 0; }```	**5 marks** **10 10**
3.	```struct Box { char *head, *mid, *tail; }_b; int main() { char *arr= "Fxam01"; char arr[] = "Fxam01"; _b.head = arr; _b.mid = arr + 2; _b.tail = arr + 3; *_b.head = 'E'; cout << "head: " << *_b.head << endl; cout << "mid: " << _b.mid << endl; cout << "tail: " << _b.tail << endl; return 0; }```	**5 marks** head: E mid: am01 tail: m01
4.	```struct Node { int data; Node *next; }; void fun(Node *head) { if (head == nullptr) return; fun(head->next); if (head->next != nullptr) head->data += head->next->data; } void display(Node *head) { if (!head) return; cout << head->data << " "; display(head->next);```	**10 marks** **6 5 3**

	`` } int main() { Node n1, n2, n3; n1.data = 1; n2.data = 2; n3.data = 3; n1.next = &n2; n2.next = &n3; n3.next = nullptr; fun(&n1); display(&n1); return 0; } ```	
5.	``` int main() { int a = 32, *ptr = &a; char ch = 'A', &cho = ch; cho += a; *ptr -= ch + 20; cout << a << ", " << ch << endl; return 0; } ```	**5 marks** **-85, a**
6.	``` struct Link { int id; Link* next; }; struct Module { int id; Link* link; }; int main() { Module A; A.id = 10; A.link = new Link; A.link->id = 20; A.link->next = new Link; A.link->next->id = 30; A.link->next->next = nullptr; cout << "STARTING TRACE" << endl; cout << "BEGIN-" << A.id << endl; Link* current = A.link; while(current != nullptr) { cout << " LINK-" << current->id << endl; current = current->next; } cout << "END-" << A.id << endl; current = A.link; while(current != nullptr) { Link* temp = current; current = current->next; delete temp; ```	**5 marks** **STARTING TRACE** **BEGIN-10** **LINK-20** **LINK-30** **END-10**

<table>
<tr><td></td><td><pre> }
  return 0;
}</pre></td><td></td></tr>
<tr><td>7.</td><td><pre>struct Sample {
   int *ptr;
   int value;
};

void update(Sample s) {
   *(s.ptr) += 5;
   s.value += 10;
}

int main() {
   int x = 10;
   Sample s1;
   s1.ptr = &x;
   s1.value = 20;

   update(s1);

   cout << x << " " << s1.value << endl;
   return 0;
}</pre></td><td>5 marks

15 20</td></tr>
<tr><td>8.</td><td><pre>void modify(int *arr, int size, int index) {
    if (index >= size)
        return;

     modify(arr, size, index + 1);

     if (index % 2 == 0)
         arr[index] += arr[size - index - 1];
}

int main() {
   int arr[] = {1,2,3,4,5};
   modify(arr, 5, 0);

   for(int i=0;i<5;i++)
       cout<<arr[i]<<" ";

   return 0;
}</pre></td><td>5 marks

7 2 6 4 6</td></tr>
<tr><td>9.</td><td><pre>int** modifyPtr(int** &p) {
   return p;
}

int main() {
  int* x = new int;
  *x=42;
  int** y = &x;
  int** z = y;
  cout << **y << endl;</pre></td><td>5 marks

42

1</td></tr>
</table>

	`` delete *z; *modifyPtr(y) = nullptr; cout << (x == nullptr) << endl; return 0; } ```	
10.	struct Data { int* p; int size; }; void fun(int n, Data d, int index) { if (n <= 1 || index >= d.size) return; d.p[index] += 10; cout << "n = " << n << " index = " << index << " value = " << d.p[index] << endl; fun(n - 2, d, index + 1); } int main() { int size = 5; int* arr = new int[size]; for (int i = 0; i < size; i++) arr[i] = 100 + i; Data d1; d1.p = arr; d1.size = size; cout << "Starting array values:" << endl; for (int i = 0; i < size; i++) cout << arr[i] << " "; cout << endl << endl; fun(6, d1, 0); cout << "\nFinal array values:" << endl; for (int i = 0; i < size; i++) cout << arr[i] << " "; cout << endl; delete[] arr; return 0; }	**10 marks** **Starting array values:** **100 101 102 103 104** **n = 6 index = 0 value = 110** **n = 4 index = 1 value = 111** **n = 2 index = 2 value = 112** **Final array values:** **110 111 112 103 104**

	Q-1
Obtained Marks	
Total Marks	**60**

Roll Number: ________________________